

Course: B. Tech

Semester: III

Branch: Electronics and Telecommunication Engg. / Electronics and Communication Engineering

Subject Code & Name: BTES304 Electrical Machines and Instruments

Max Marks: 60

Date: 12/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 is compulsory.
3. Attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

				(Level/CO)	Marks
Q.1 Objective type questions. (Compulsory Question)					12
1	Which part of a DC machine is responsible for generating EMF? a. Stator b. Rotor c. Commutator d. Armature			Level 1	1
2	In a DC generator, the role of the commutator is to: a. Convert AC to DC b. Convert DC to AC c. Increase voltage d. Reduce losses			Level 2	1
3	If the speed of a DC generator is doubled, the generated EMF will: a. Remain constant b. Double c. Reduce by half d. Increase four times			Level 3	1
4	Which DC motor has the best speed regulation? a. Series motor b. Shunt motor c. Compound motor d. None of the above			Level 2	1
5	If the load on a DC motor increases, the back EMF: a. Increases b. Decreases c. Remains constant d. Becomes zero			Level 3	1
6	A three-point starter is generally used for: a. DC series motor b. DC shunt motor c. DC compound motor d. All of the above			Level 1	1
7	Increasing the field resistance in a DC shunt motor results in: a. Increase in speed b. Decrease in speed c. No effect on speed d. Increase in current			Level 3	1
8	The slip of an induction motor at no load is approximately: a. 100% b. 0% c. 50% d. 2%			Level 1	1
9	The main working principle of a hysteresis motor is based on a. Induction heating b. Residual magnetism c. Variable reluctance d. Eddy currents			Level 2	1

10	Hall Effect transducers are primarily used for measuring: a. Temperature b. Pressure c. Magnetic field d. Displacement	Level 3	1
11	The output of an LVDT is: a. Digital b. AC voltage c. DC voltage d. Resistance change	Level 1	1
12	Which of the following instruments is used for measuring rotational speed in RPM a. Sound level meter b. Tachometer c. Optical oscillograph d. X-Y plotter	Level 1	1
Q.2	Solve the following.		12
A)	With a neat diagram, explain the construction and working of a DC generator.	Level 2	6
B)	What are the different methods of speed control in a DC motor? Explain any two.	Level 3	6
Q.3	Solve the following.		12
A)	Describe the construction, working principle, and types of synchronous motors with neat diagrams.	Level 2	6
B)	Explain the construction and working principle of a three-phase induction motor. Also Derive the torque equation of an induction motor.	Level 3	6
Q.4	Solve Any Two of the following.		12
A)	Explain the construction and working of a servo motor. How is it controlled in automation systems?	Level 3	6
B)	Explain the hysteresis motor in detail. Why is it preferred in applications requiring noiseless operation?	Level 2	6
C)	Explain in detail the construction, working, and applications of a stepper motor.	Level 2	6
Q.5	Solve Any Two of the following.		12
A)	What are the different types of temperature transducers? Explain any two in detail.	Level 3	6
B)	Describe the working principle and applications of strain gauges.	Level 3	6
C)	Explain the working principle of proximity sensors and classify them.	Level 2	6
Q.6	Solve Any Two of the following.		12
A)	Discuss the principle and applications of X-Y plotters and optical oscillograph	Level 2	6
B)	Explain the methods of vibration measurement and their industrial applications.	Level 3	6
C)	What is a VAW meter? Explain its working and applications in power systems.	Level 3	6

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